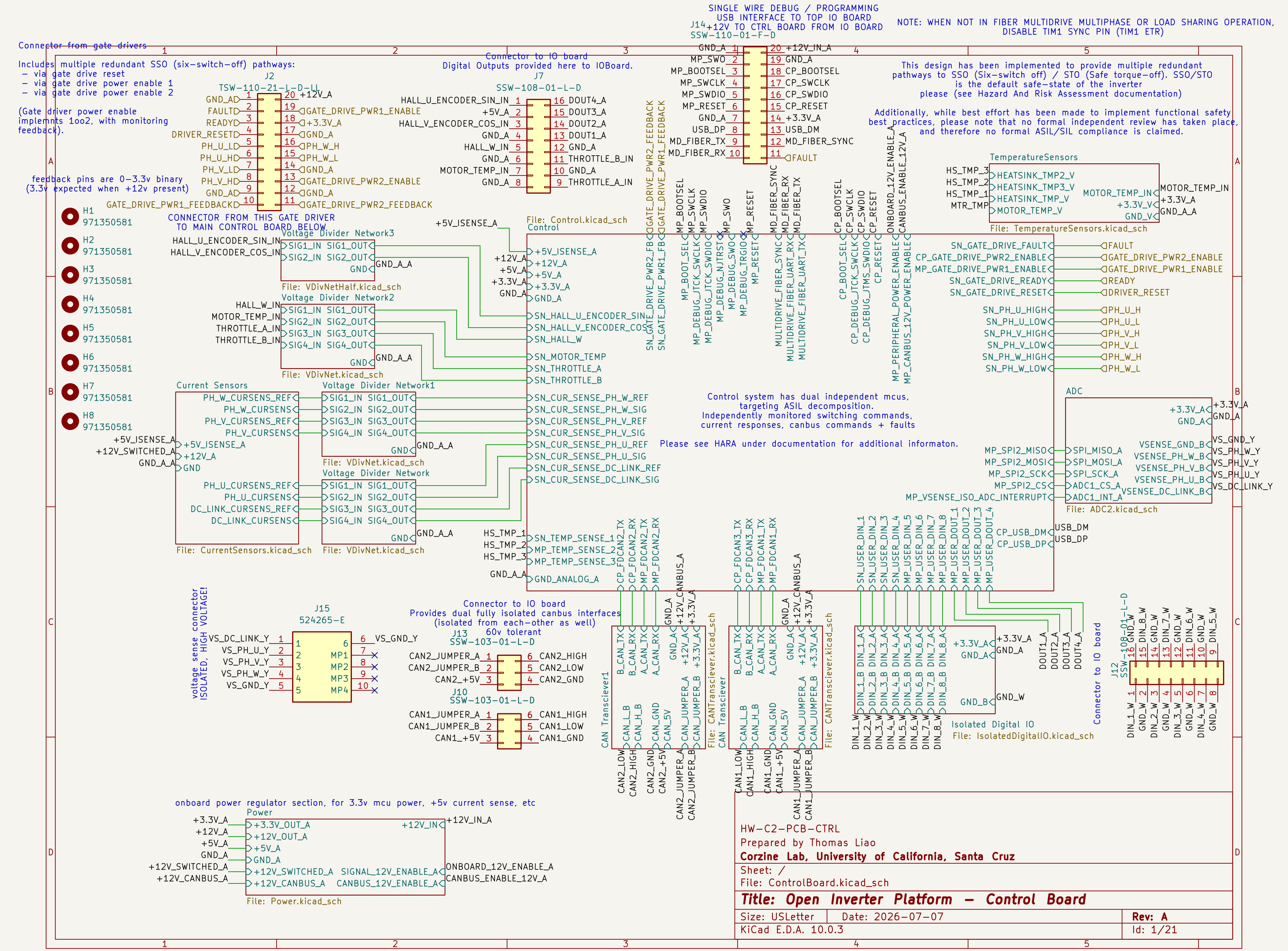
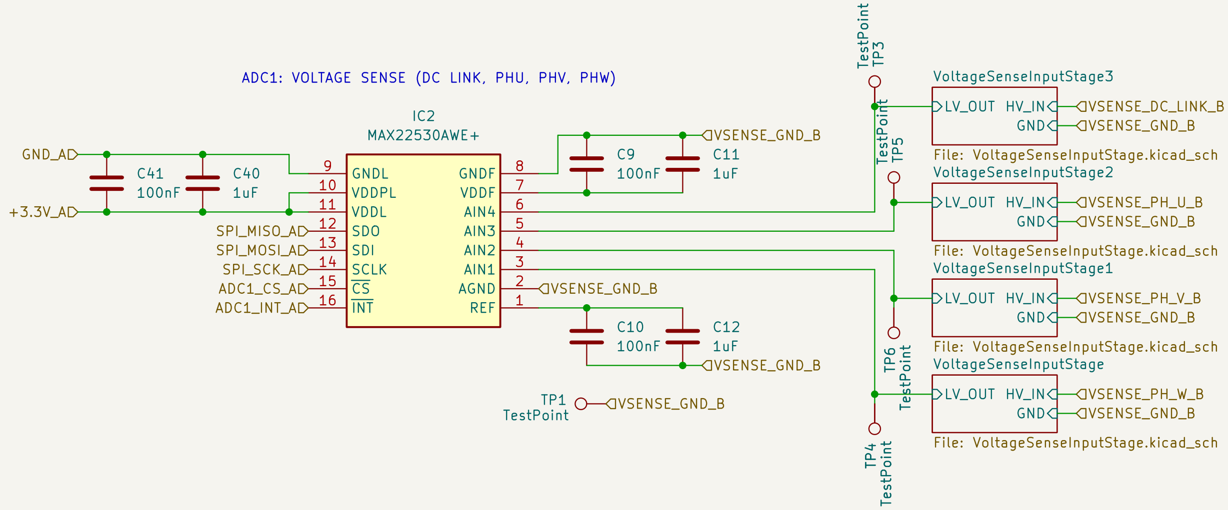


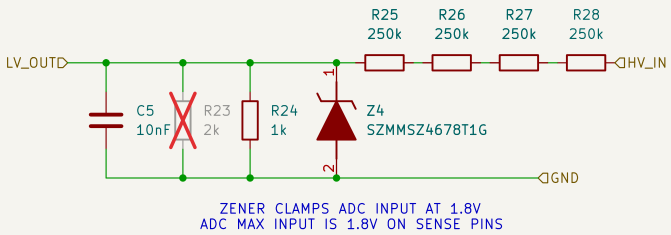
Connector from gate drivers
Includes multiple redundant SSO (six-switch-off) pathways:
- via gate drive reset
- via gate drive power enable 1
- via gate drive power enable 2
(Gate driver power enable implemnts 1oo2, with monitoring feedback).
feedback pins are 0-3.3v binary (3.3v expected when +12v present)
J2
TSW-110-21-L-D-LL
GND_A 1
FAULT 2
READY 3
DRIVER_RESET 4
PH_U_L 5
PH_U_H 6
PH_V_L 7
PH_V_H 8
GND_A 9
GATE_DRIVE_PWR1_FEEDBACK 10
20 +12V_A
19 GATE_DRIVE_PWR1_ENABLE
18 +3.3V_A
17 GND_A
16 PH_W_H
15 PH_W_L
14 GND_A
13 GATE_DRIVE_PWR2_ENABLE
12 GND_A
11 GATE_DRIVE_PWR2_FEEDBACK
CONNECTOR FROM THIS GATE DRIVER TO MAIN CONTROL BOARD BELOW
H1 971350581
H2 971350581
H3 971350581
H4 971350581
H5 971350581
H6 971350581
H7 971350581
H8 971350581
Connector to IO board
Digital Outputs provided here to IOBoard.
J7
SSW-108-01-L-D
HALL_U_ENCODER_SIN_IN 1
+5V_A 2
HALL_V_ENCODER_COS_IN 3
GND_A 4
HALL_W_IN 5
GND_A 6
MOTOR_TEMP_IN 7
GND_A 8
16 DOUT4_A
15 DOUT3_A
14 DOUT2_A
13 DOUT1_A
12 GND_A
11 THROTTLE_B_IN
10 GND_A
9 THROTTLE_A_IN
SINGLE WIRE DEBUG / PROGRAMMING
USB INTERFACE TO TOP IO BOARD
+12V TO CTRL BOARD FROM IO BOARD
J14
SSW-110-01-F-D
GND_A 1
MP_SWO 2
MP_BOOTSEL 3
MP_SWCLK 4
MP_SWDIO 5
MP_RESET 6
GND_A 7
USB_DP 8
MD_FIBER_TX 9
MD_FIBER_RX 10
20 +12V_IN_A
19 GND_A
18 CP_BOOTSEL
17 CP_SWCLK
16 CP_SWDIO
15 CP_RESET
14 +3.3V_A
13 USB_DM
12 MD_FIBER_SYNC
11 FAULT
NOTE: WHEN NOT IN FIBER MULTIDRIVE MULTIPHASE OR LOAD SHARING OPERATION, DISABLE TIM1 SYNC PIN (TIM1 ETR)
This design has been implemented to provide multiple redundant pathways to SSO (Six-switch off) / STO (Safe torque-off). SSO/STO is the default safe-state of the inverter please (see Hazard And Risk Assessment documentation)
Additionally, while best effort has been made to implement functional safety best practices, please note that no formal independent review has taken place, and therefore no formal ASIL/SIL compliance is claimed.
TemperatureSensors
HS_TMP_3 HEATSINK_TMP2_V
HS_TMP_2 HEATSINK_TMP3_V
HS_TMP_1 HEATSINK_TMP_V
MTR_TMP MOTOR_TEMP_V
MOTOR_TEMP_IN MOTOR_TEMP_IN
+3.3V_V +3.3V_A
GND_V GND_A_A
File: TemperatureSensors.kicad_sch
Voltage Divider Network3
HALL_U_ENCODER_SIN_IN SIG1_IN SIG1_OUT
HALL_V_ENCODER_COS_IN SIG2_IN SIG2_OUT
GND GND_A_A
File: VDivNetHalf.kicad_sch
Voltage Divider Network2
HALL_W_IN SIG1_IN SIG1_OUT
MOTOR_TEMP_IN SIG2_IN SIG2_OUT
THROTTLE_A_IN SIG3_IN SIG3_OUT
THROTTLE_B_IN SIG4_IN SIG4_OUT
GND GND_A_A
File: VDivNet.kicad_sch
Current Sensors
PH_W_CURSENS_REF
PH_W_CURSENS
PH_V_CURSENS_REF
PH_V_CURSENS
+5V_ISENSE_A +5V_ISENSE_A
+12V_SWITCHED_A +12V_A
GND_A_A GND
PH_U_CURSENS_REF
PH_U_CURSENS
DC_LINK_CURSENS_REF
DC_LINK_CURSENS
File: CurrentSensors.kicad_sch
Voltage Divider Network1
SIG1_IN SIG1_OUT
SIG2_IN SIG2_OUT
SIG3_IN SIG3_OUT
SIG4_IN SIG4_OUT
GND GND_A_A
File: VDivNet.kicad_sch
Voltage Divider Network
SIG1_IN SIG1_OUT
SIG2_IN SIG2_OUT
SIG3_IN SIG3_OUT
SIG4_IN SIG4_OUT
GND GND_A_A
File: VDivNet.kicad_sch
File: Control.kicad_sch
Control
+5V_ISENSE_A +5V_ISENSE_A
+12V_A +12V_A
+5V_A +5V_A
+3.3V_A +3.3V_A
GND_A GND_A
SN_HALL_U_ENCODER_SIN
SN_HALL_V_ENCODER_COS
SN_HALL_W
SN_MOTOR_TEMP
SN_THROTTLE_A
SN_THROTTLE_B
SN_CUR_SENSE_PH_W_REF
SN_CUR_SENSE_PH_W_SIG
SN_CUR_SENSE_PH_V_REF
SN_CUR_SENSE_PH_V_SIG
SN_CUR_SENSE_PH_U_REF
SN_CUR_SENSE_PH_U_SIG
SN_CUR_SENSE_DC_LINK_REF
SN_CUR_SENSE_DC_LINK_SIG
HS_TMP_1 SN_TEMP_SENSE_1
HS_TMP_2 MP_TEMP_SENSE_2
HS_TMP_3 MP_TEMP_SENSE_3
GND_A_A GND_ANALOG_A
GATE_DRIVE_PWR2_FEEDBACK SN_GATE_DRIVE_PWR2_FB
GATE_DRIVE_PWR1_FEEDBACK SN_GATE_DRIVE_PWR1_FB
MP_BOOTSEL MP_BOOT_SEL
MP_SWCLK MP_DEBUG_JTCK_SWCLK
MP_SWDIO MP_DEBUG_JTCK_SWDIO
MP_DEBUG_NJTRST
MP_SWO MP_DEBUG_SWO
MP_DEBUG_TRGIO
MP_RESET MP_RESET
MD_FIBER_SYNC MULTIDRIVE_FIBER_SYNC
MD_FIBER_RX MULTIDRIVE_FIBER_UART_RX
MD_FIBER_TX MULTIDRIVE_FIBER_UART_TX
CP_BOOTSEL CP_BOOT_SEL
CP_SWCLK CP_DEBUG_JTCK_SWCLK
CP_SWDIO CP_DEBUG_JTMS_SWDIO
CP_RESET CP_RESET
ONBOARD_12V_ENABLE_A MP_PERIPHERAL_POWER_ENABLE
CANBUS_ENABLE_12V_A MP_CANBUS_12V_POWER_ENABLE
SN_GATE_DRIVE_FAULT FAULT
CP_GATE_DRIVE_PWR2_ENABLE GATE_DRIVE_PWR2_ENABLE
MP_GATE_DRIVE_PWR1_ENABLE GATE_DRIVE_PWR1_ENABLE
SN_GATE_DRIVE_READY READY
SN_GATE_DRIVE_RESET DRIVER_RESET
SN_PH_U_HIGH PH_U_H
SN_PH_U_LOW PH_U_L
SN_PH_V_HIGH PH_V_H
SN_PH_V_LOW PH_V_L
SN_PH_W_HIGH PH_W_H
SN_PH_W_LOW PH_W_L
Control system has dual independent mcus, targeting ASIL decomposition. Independently monitored switching commands, current responses, canbus commands + faults
Please see HARA under documentation for additional informaton.
MP_SPI2_MISO
MP_SPI2_MOSI
MP_SPI2_SCK
MP_SPI2_CS
MP_VSENSE_ISO_ADC_INTERRUPT
CP_USB_DM USB_DM
CP_USB_DP USB_DP
CP_FDCAN2_TX
CP_FDCAN2_RX
MP_FDCAN2_TX
MP_FDCAN2_RX
GND_A
+12V_CANBUS_A
+3.3V_A
CP_FDCAN3_TX
CP_FDCAN3_RX
MP_FDCAN1_TX
MP_FDCAN1_RX
GND_A
+12V_CANBUS_A
+3.3V_A
SN_USER_DIN_1
SN_USER_DIN_2
SN_USER_DIN_3
SN_USER_DIN_4
MP_USER_DIN_5
MP_USER_DIN_6
MP_USER_DIN_7
MP_USER_DIN_8
MP_USER_DOUT_1
MP_USER_DOUT_2
MP_USER_DOUT_3
MP_USER_DOUT_4
ADC
+3.3V_A +3.3V_A
GND_A GND_A
SPI_MISO_A
SPI_MOSI_A
SPI_SCK_A
ADC1_CS_A
ADC1_INT_A
VSENSE_GND_B VS_GND_Y
VSENSE_PH_W_B VS_PH_W_Y
VSENSE_PH_V_B VS_PH_V_Y
VSENSE_PH_U_B VS_PH_U_Y
VSENSE_DC_LINK_B VS_DC_LINK_Y
File: ADC2.kicad_sch
voltage sense connector
ISOLATED, HIGH VOLTAGE!
J15
524265-E
VS_DC_LINK_Y 1
VS_PH_U_Y 2
VS_PH_V_Y 3
VS_PH_W_Y 4
VS_GND_Y 5
6 VS_GND_Y
MP1 7
MP2 8
MP3 9
MP4 10
Connector to IO board
Provides dual fully isolated canbus interfaces (isolated from each-other as well)
60v tolerant
J13
SSW-103-01-L-D
CAN2_JUMPER_A 1
CAN2_JUMPER_B 2
CAN2_+5V 3
6 CAN2_HIGH
5 CAN2_LOW
4 CAN2_GND
J10
SSW-103-01-L-D
CAN1_JUMPER_A 1
CAN1_JUMPER_B 2
CAN1_+5V 3
6 CAN1_HIGH
5 CAN1_LOW
4 CAN1_GND
CAN Transciever1
B_CAN_TX
B_CAN_RX
A_CAN_TX
A_CAN_RX
GND_A
+12V_A
+3.3V_A
CAN_L_B
CAN_H_B
CAN_GND
CAN_5V
CAN_JUMPER_A
CAN_JUMPER_B
File: CANTransciever.kicad_sch
CAN2_LOW
CAN2_HIGH
CAN2_GND
CAN2_+5V
CAN2_JUMPER_A
CAN2_JUMPER_B
CAN Transciever
B_CAN_TX
B_CAN_RX
A_CAN_TX
A_CAN_RX
GND_A
+12V_A
+3.3V_A
CAN_L_B
CAN_H_B
CAN_GND
CAN_5V
CAN_JUMPER_A
CAN_JUMPER_B
File: CANTransciever.kicad_sch
CAN1_LOW
CAN1_HIGH
CAN1_GND
CAN1_+5V
CAN1_JUMPER_A
CAN1_JUMPER_B
DIN_1_A DIN_1_B
DIN_2_A DIN_2_B
DIN_3_A DIN_3_B
DIN_4_A DIN_4_B
DIN_5_A DIN_5_B
DIN_6_A DIN_6_B
DIN_7_A DIN_7_B
DIN_8_A DIN_8_B
+3.3V_A +3.3V_A
GND_A GND_A
GND_B GND_W
Isolated Digital IO
File: IsolatedDigitalIO.kicad_sch
DIN_1_W
DIN_2_W
DIN_3_W
DIN_4_W
DIN_5_W
DIN_6_W
DIN_7_W
DIN_8_W
DOUT1_A
DOUT2_A
DOUT3_A
DOUT4_A
Connector to IO board
J12
SSW-108-01-L-D
1 DIN_1_W
2 GND_W
3 DIN_2_W
4 GND_W
5 DIN_3_W
6 GND_W
7 DIN_4_W
8 GND_W
16 GND_W
15 DIN_8_W
14 GND_W
13 DIN_7_W
12 GND_W
11 DIN_6_W
10 GND_W
9 DIN_5_W
onboard power regulator section, for 3.3v mcu power, +5v current sense, etc
Power
+3.3V_A +3.3V_OUT_A
+12V_A +12V_OUT_A
+5V_A +5V_A
GND_A GND_A
+12V_SWITCHED_A +12V_SWITCHED_A
+12V_CANBUS_A +12V_CANBUS_A
+12V_IN +12V_IN_A
SIGNAL_12V_ENABLE_A ONBOARD_12V_ENABLE_A
CANBUS_12V_ENABLE_A CANBUS_ENABLE_12V_A
File: Power.kicad_sch
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /
File: ControlBoard.kicad_sch
Title: Open Inverter Platform - Control Board
Size: USLetter
Date: 2026-07-07
KiCad E.D.A. 10.0.3
Rev: A
Id: 1/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/
File: ADC2.kicad_sch

Title: Open Inverter Platform – Control Board

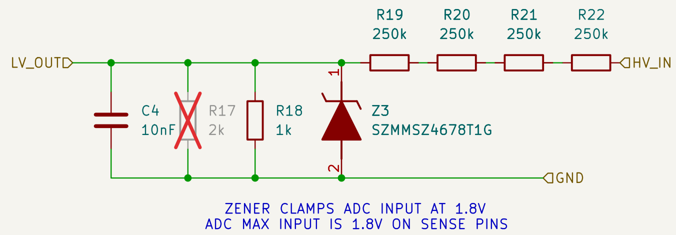
Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 4/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform – Control Board

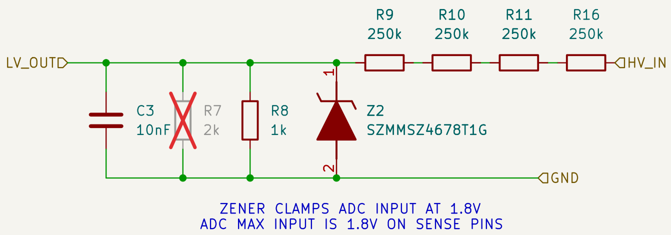
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 2/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage1/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform – Control Board

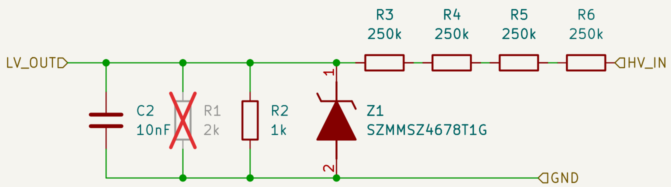
Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 3/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage2/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 5/21

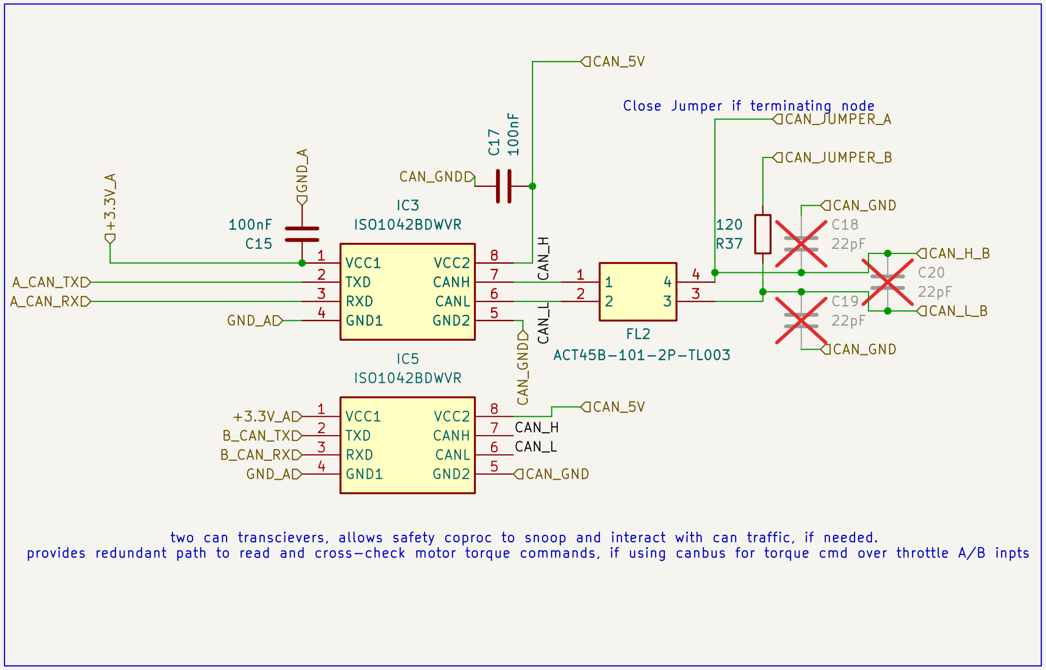


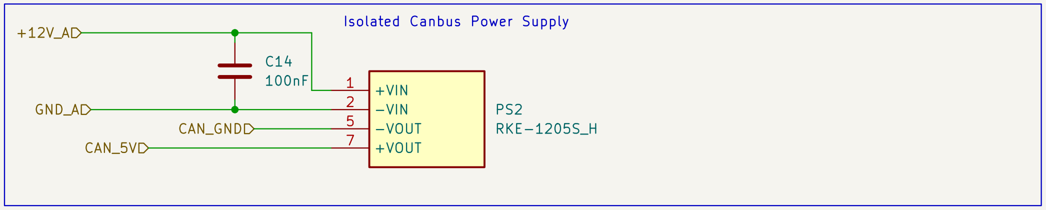
ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /ADC/VoltageSenseInputStage3/
File: VoltageSenseInputStage.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 6/21

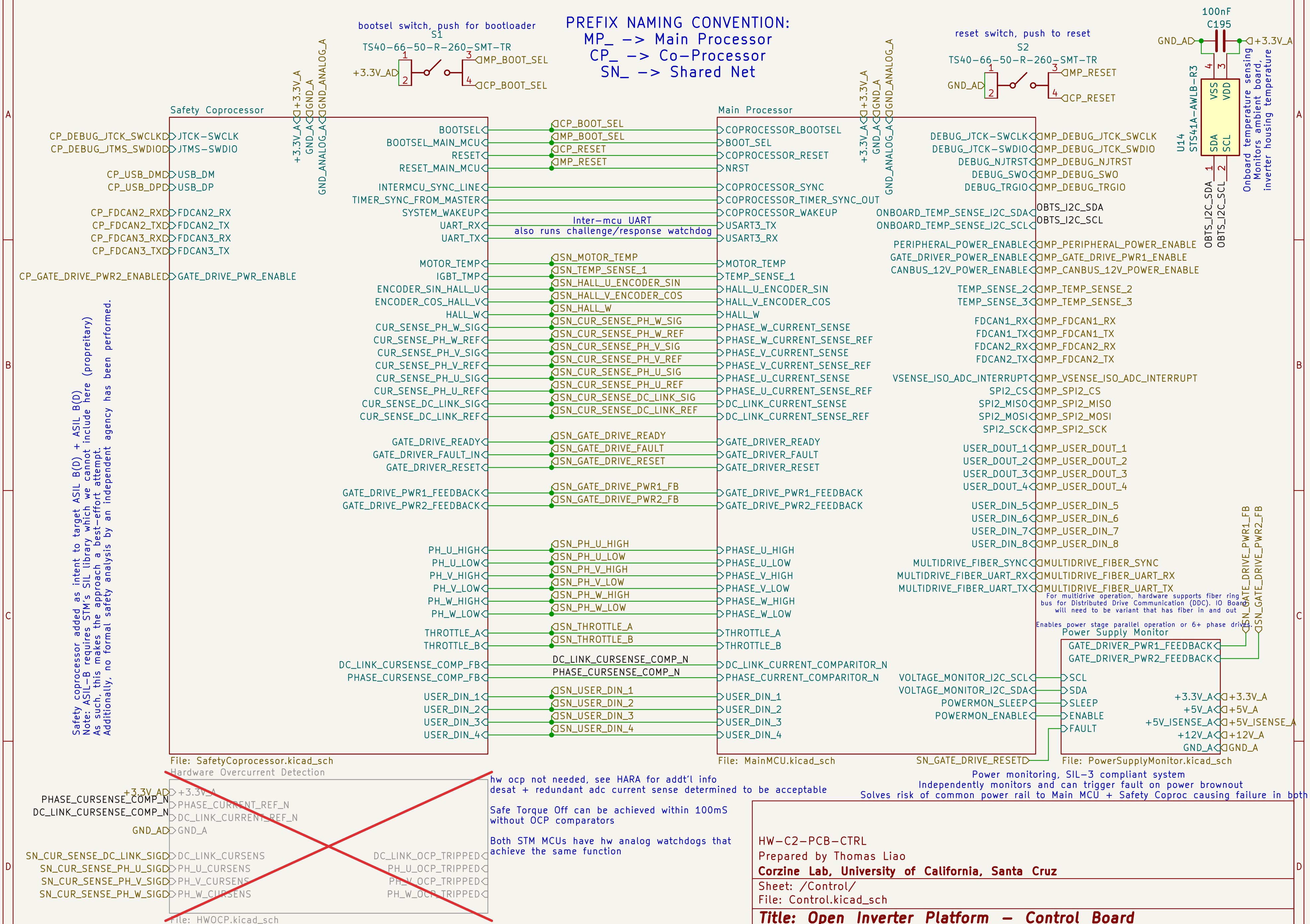




HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /CAN Transciever1/
File: CANTransciever.kicad_sch

Title: Open Inverter Platform - Control Board

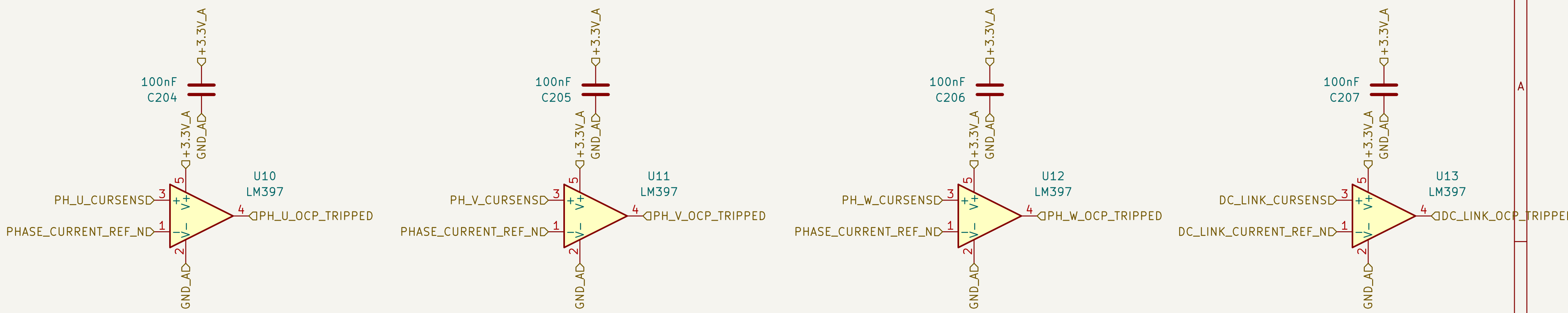
Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 7/21



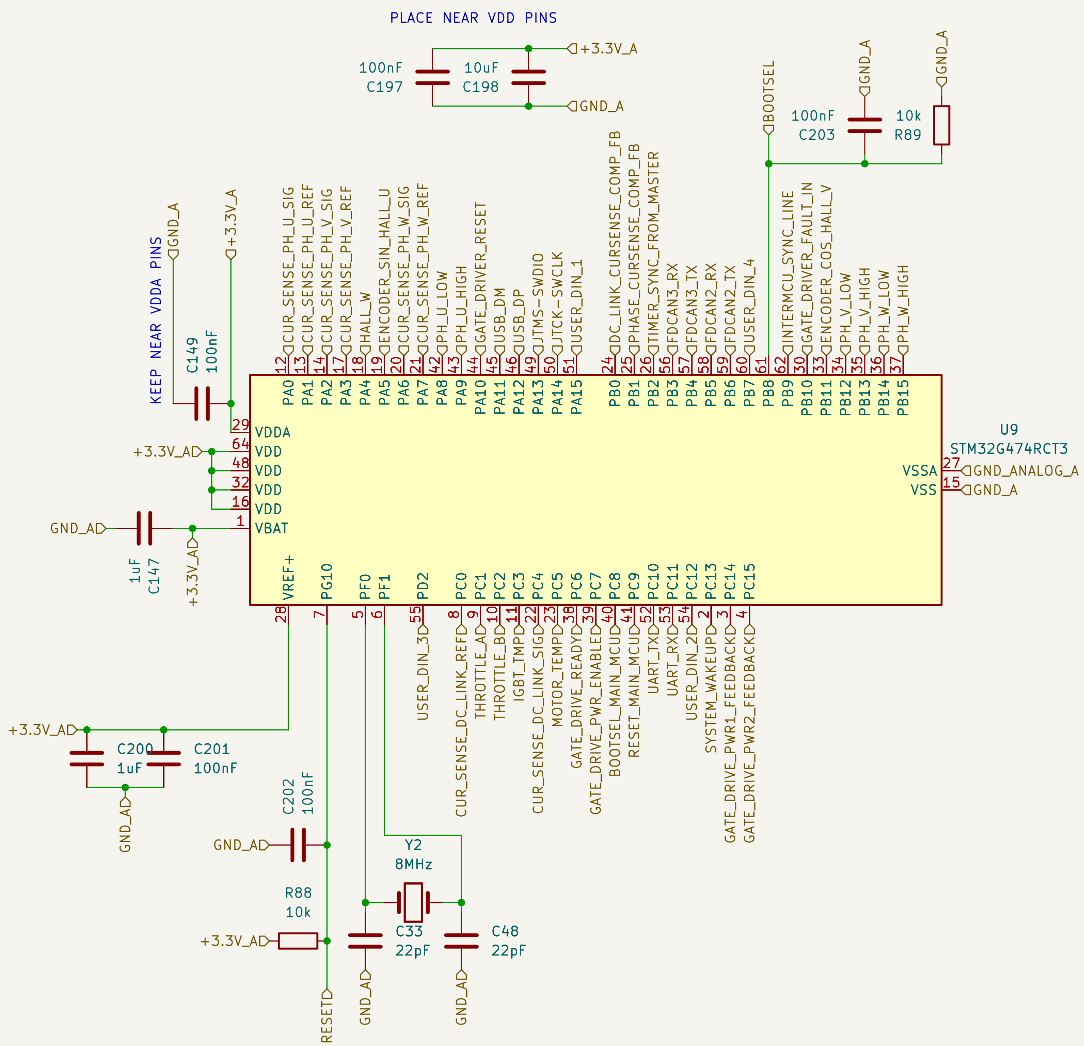
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/
File: Control.kicad_sch

Title: Open Inverter Platform - Control Board

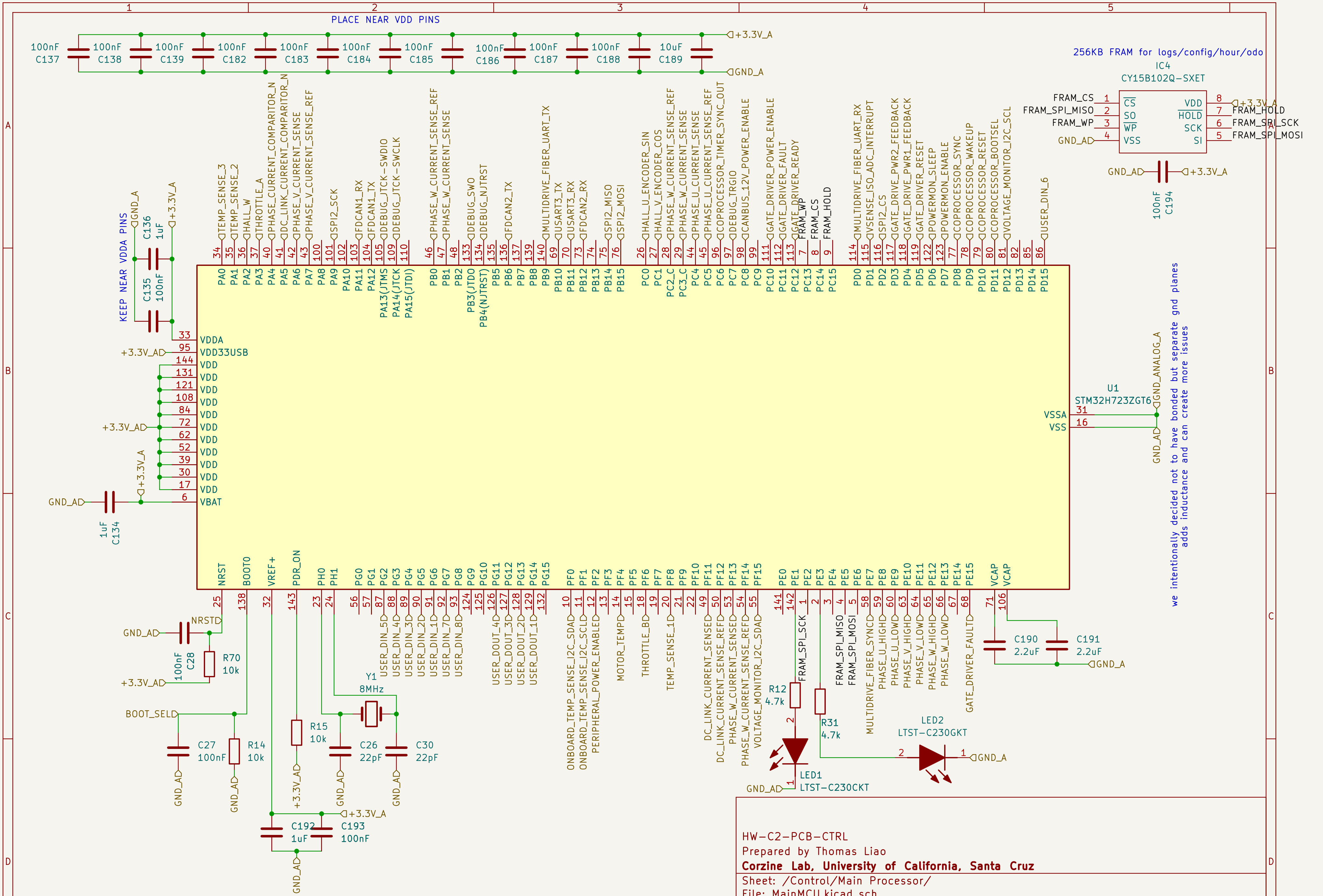
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 8/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Hardware Overcurrent Detection/
File: HWOCP.kicad_sch
Title: Open Inverter Platform - Control Board
Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 10/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Control/Safety Coprocessor/
File: SafetyCoprocessor.kicad_sch
Title: Open Inverter Platform – Control Board
Size: USLetter Date: 2026-07-07 **Rev: A**
KiCad E.D.A. 10.0.3 Id: 16/21



HW-C2-PCB-CTRL

Prepared by Thomas Liao

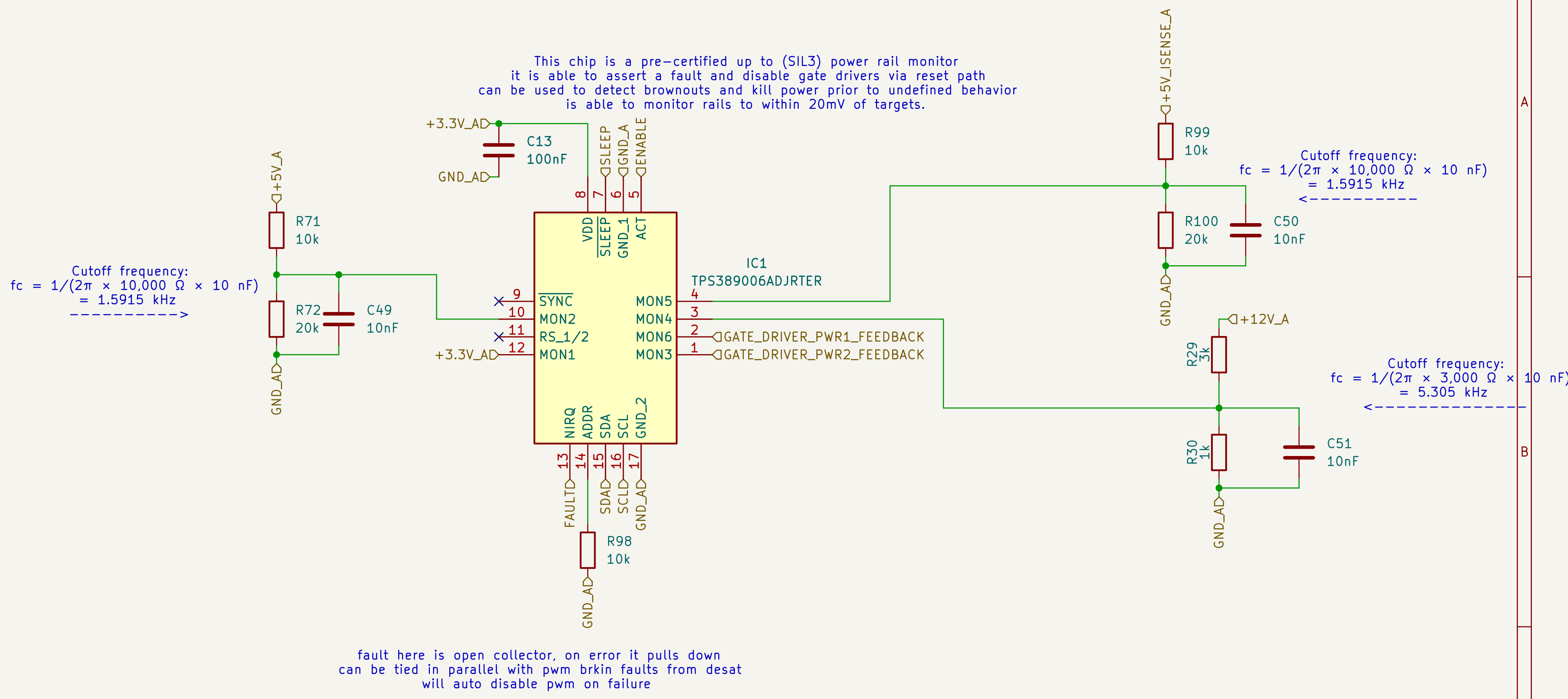
Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Main Processor/

File: MainMCU.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 19/21



HW−C2−PCB−CTRL

Prepared by Thomas Liao

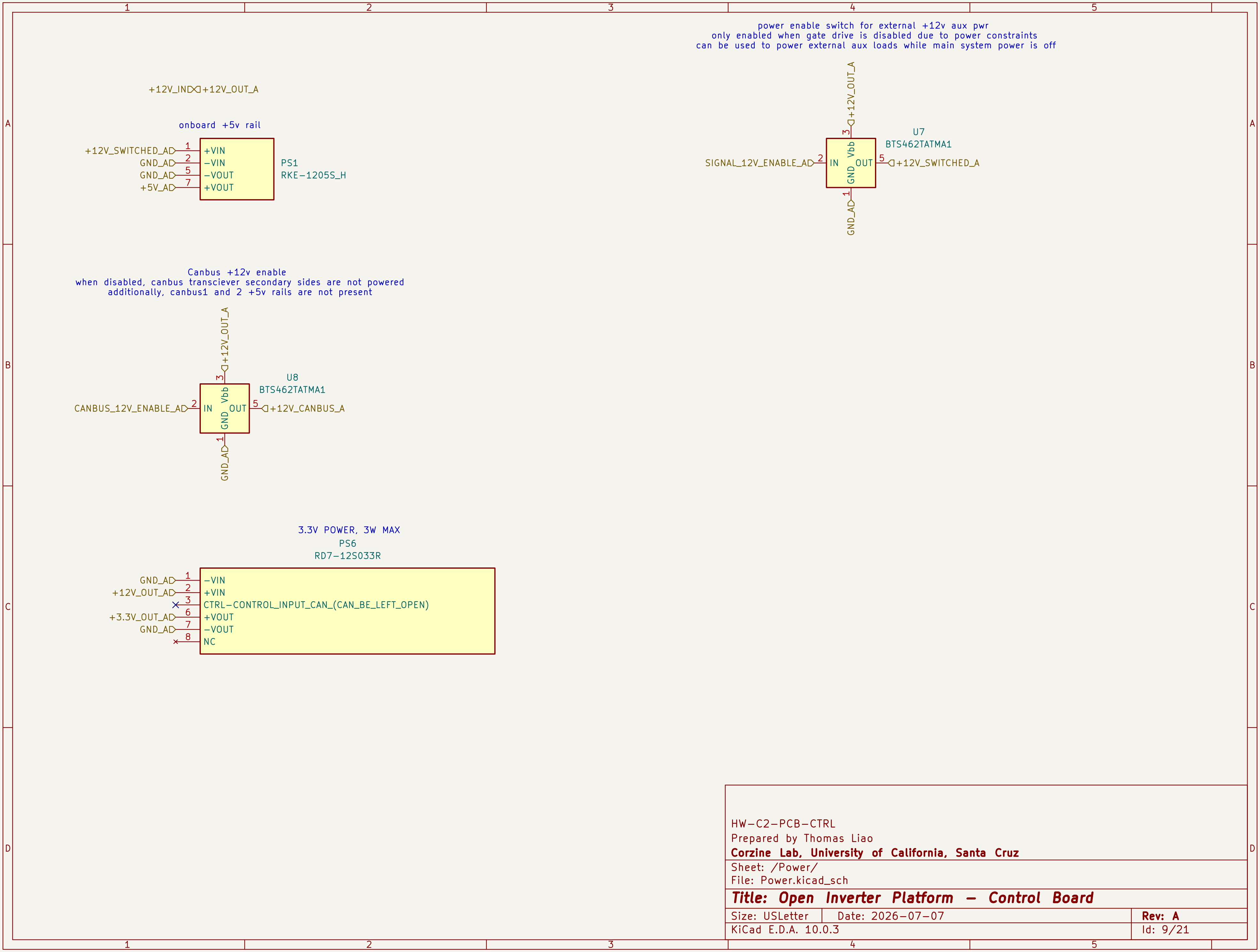
Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Power Supply Monitor/

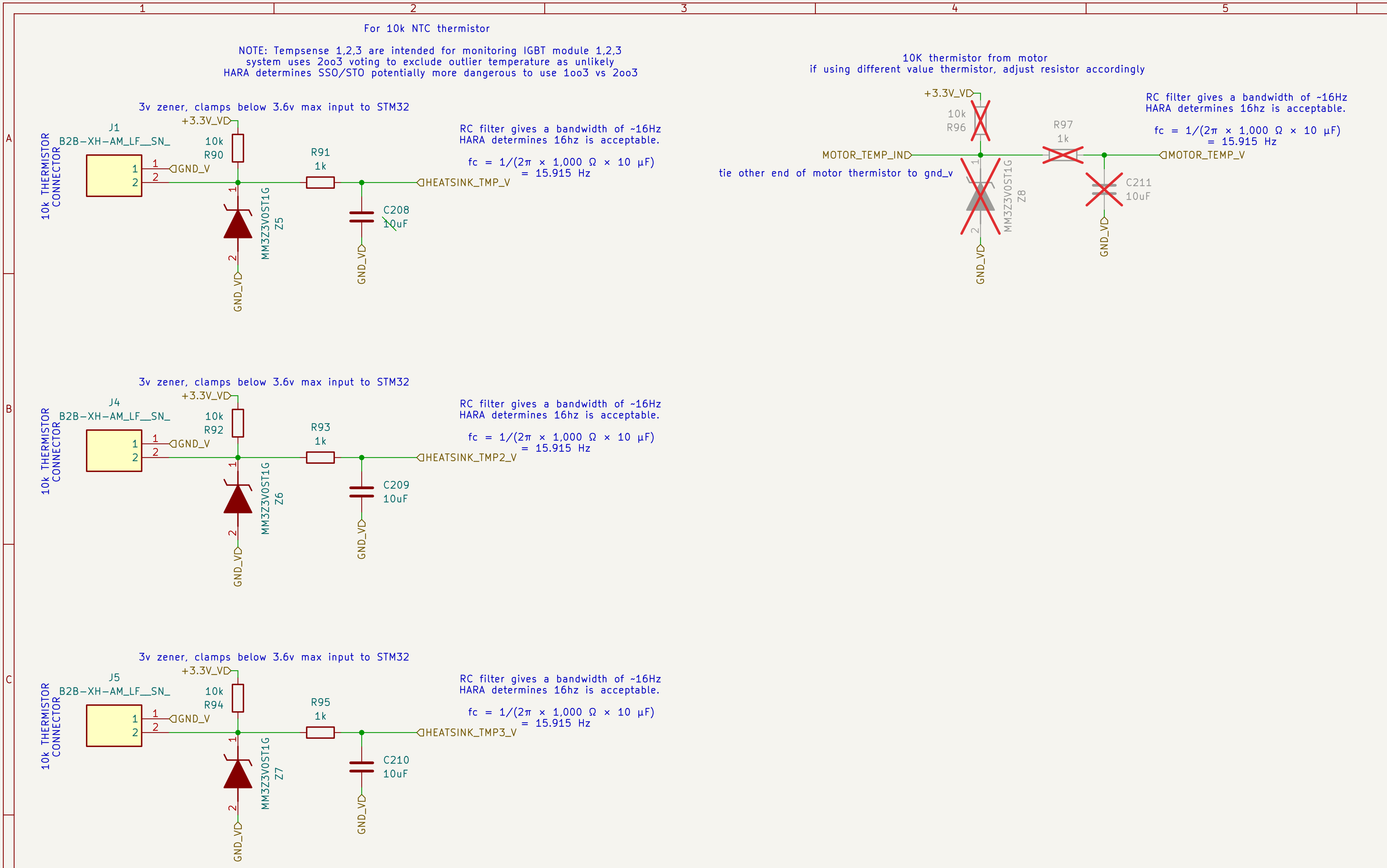
File: PowerSupplyMonitor.kicad_sch

Title: Open Inverter Platform − Control Board

Size: USLetter	Date: 2026−07−07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 20/21



power enable switch for external +12v aux pwr
only enabled when gate drive is disabled due to power constraints
can be used to power external aux loads while main system power is off
+12V_IN
+12V_OUT_A
onboard +5v rail
+12V_SWITCHED_AD
GND_AD
GND_AD
+5V_AD
+VIN
-VIN
-VOUT
+VOUT
PS1
RKE-1205S_H
+12V_OUT_A
U7
BTS462TATMA1
SIGNAL_12V_ENABLE_AD
IN
Vbb
GND
OUT
+12V_SWITCHED_A
GND_AD
Canbus +12v enable
when disabled, canbus transciever secondary sides are not powered
additionally, canbus1 and 2 +5v rails are not present
+12V_OUT_A
U8
BTS462TATMA1
CANBUS_12V_ENABLE_AD
IN
Vbb
GND
OUT
+12V_CANBUS_A
GND_AD
3.3V POWER, 3W MAX
PS6
RD7-12S033R
GND_AD
+12V_OUT_AD
+3.3V_OUT_AD
GND_AD
-VIN
+VIN
CTRL-CONTROL_INPUT_CAN_(CAN_BE_LEFT_OPEN)
+VOUT
-VOUT
NC
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Power/
File: Power.kicad_sch
Title: Open Inverter Platform - Control Board
Size: USLetter
Date: 2026-07-07
Rev: A
KiCad E.D.A. 10.0.3
Id: 9/21

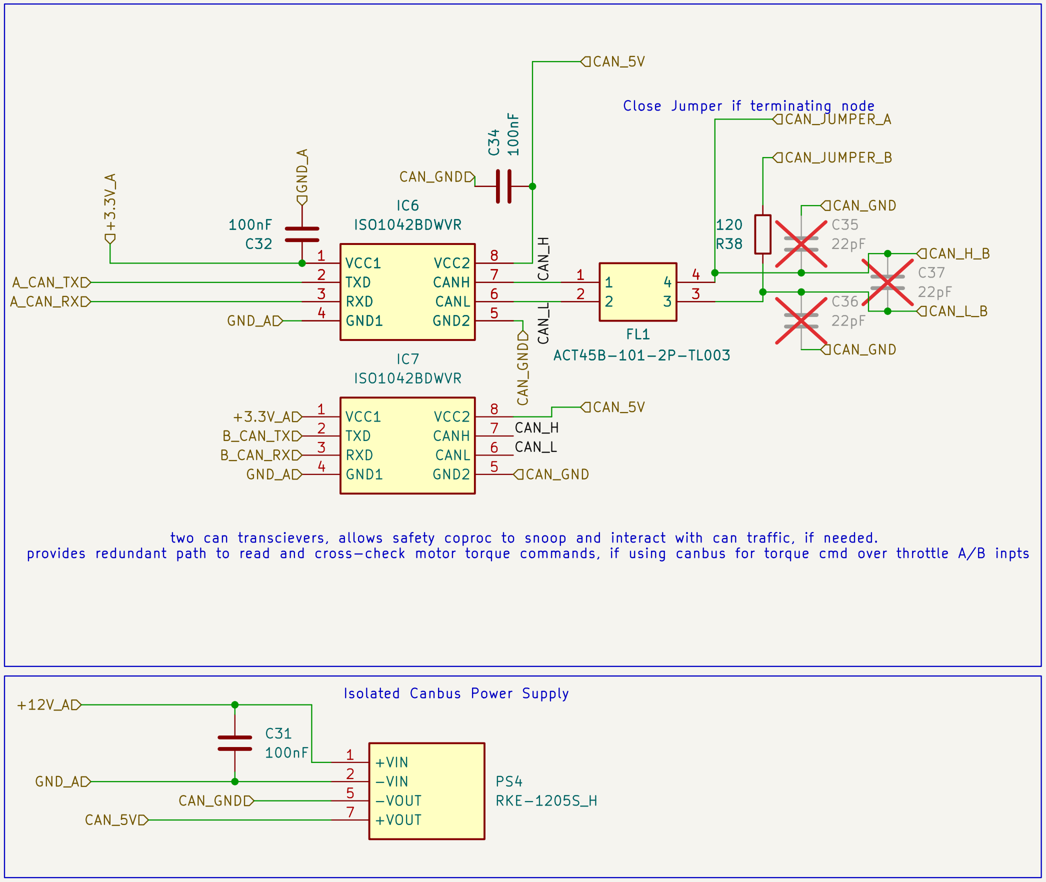


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /TemperatureSensors/
File: TemperatureSensors.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 9/21



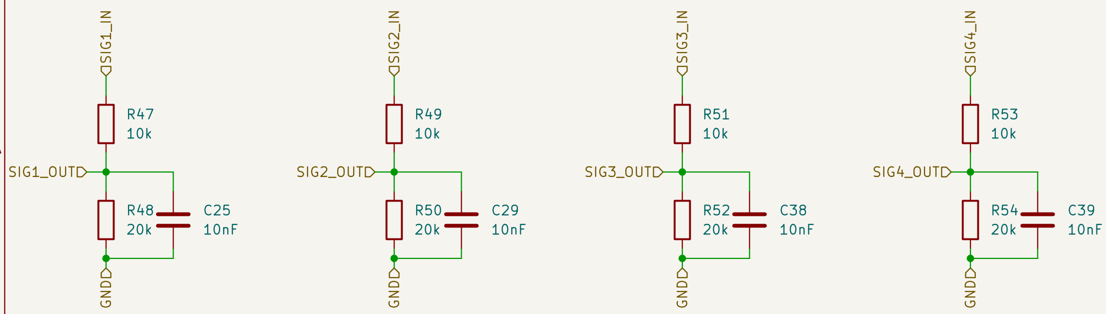
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /CAN Transciever/
File: CANTransciever.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**

KiCad E.D.A. 10.0.3 | Id: 11/21

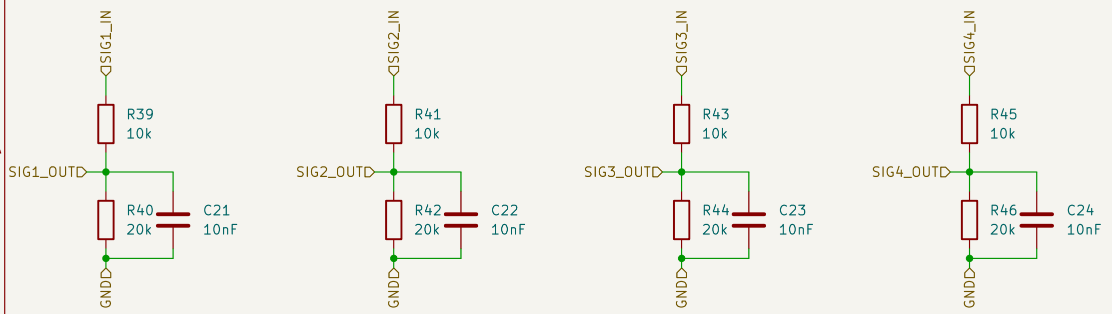


HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network1/
File: VDivNet.kicad_sch

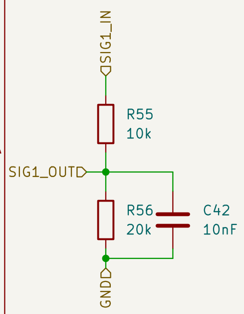
Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 12/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network/
File: VDivNet.kicad_sch
Title: Open Inverter Platform – Control Board

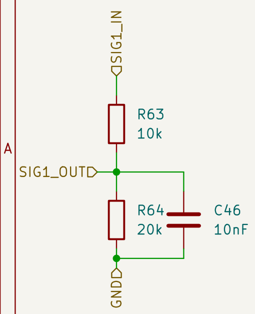
Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 13/21



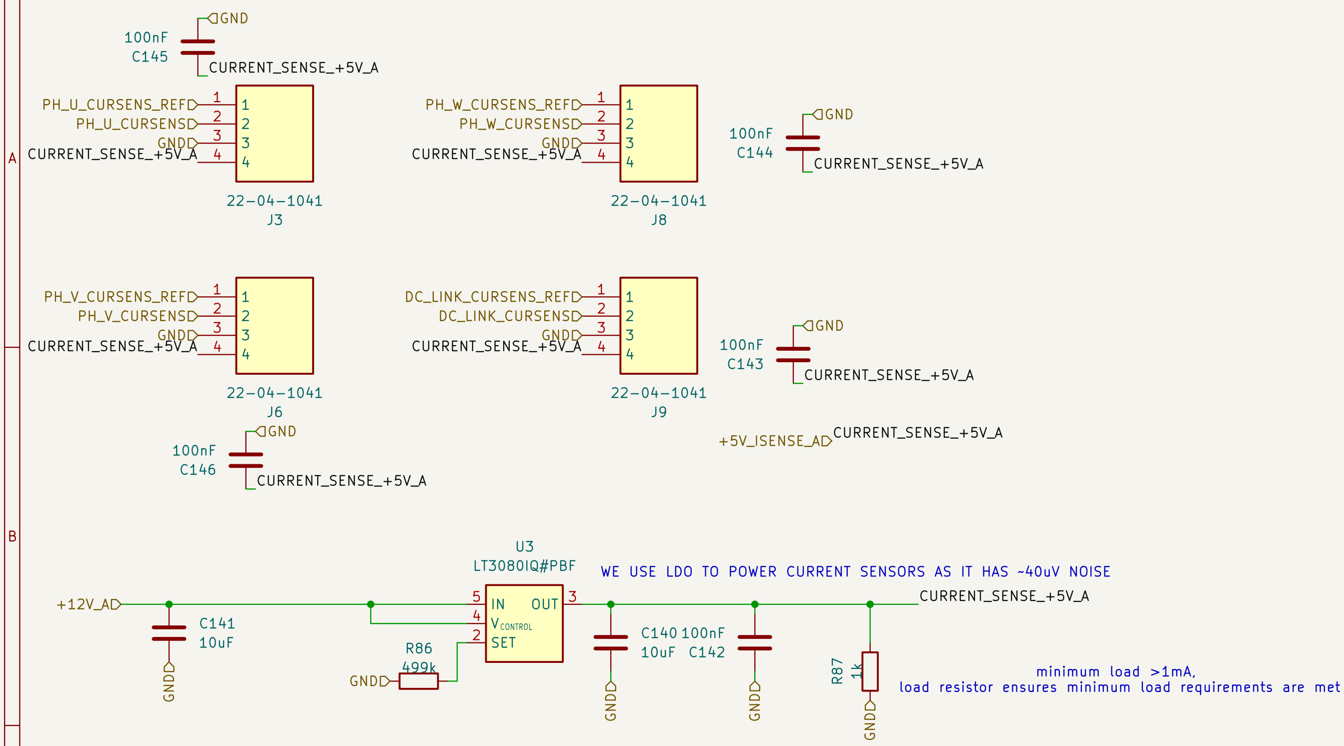
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network2/
File: VDivNet.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 14/21



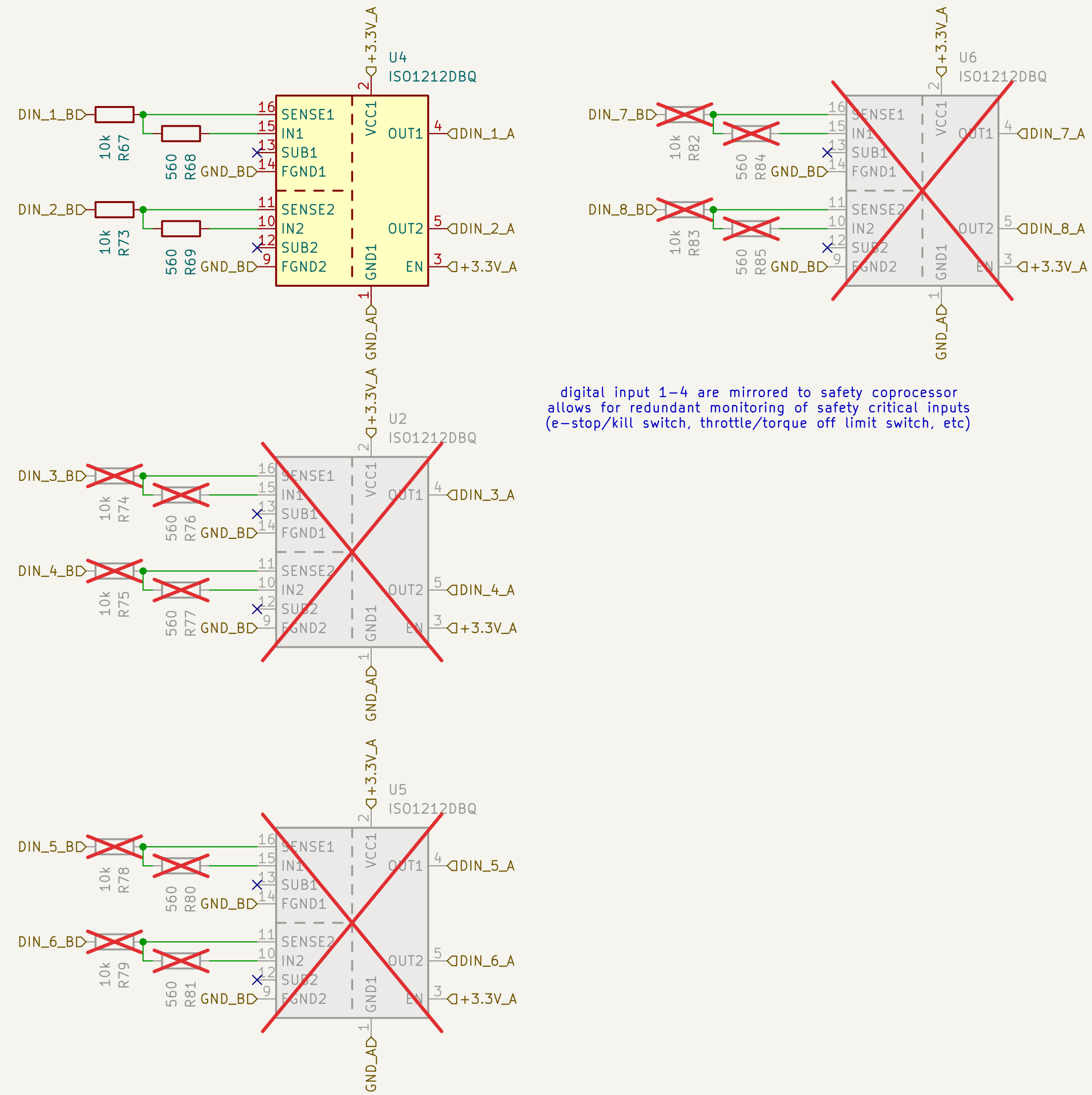
HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Voltage Divider Network3/
File: VDivNetHalf.kicad_sch
Title: Open Inverter Platform – Control Board
Size: USLetter | Date: 2026-07-07 | **Rev: A**
KiCad E.D.A. 10.0.3 | Id: 15/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Current Sensors/
File: CurrentSensors.kicad_sch

Title: Open Inverter Platform - Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 17/21



HW-C2-PCB-CTRL
Prepared by Thomas Liao
Corzine Lab, University of California, Santa Cruz
Sheet: /Isolated Digital IO/
File: IsolatedDigitalIO.kicad_sch

Title: Open Inverter Platform – Control Board

Size: USLetter	Date: 2026-07-07	**Rev: A**
KiCad E.D.A. 10.0.3		Id: 18/21